

## Lösungen Übungsblatt 1 (Potenzen, Wurzeln und Logarithmen)

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### 1. Aufgabe

Vereinfachen Sie und stellen Sie das Ergebnis mit positivem Exponenten dar:

(a)

$$(x^2y^{-3})^{-2} (x^{-2}y^3)^2 = ?$$

**Lösung:**

$$(x^2y^{-3})^{-2} (x^{-2}y^3)^2 = x^{-4}y^6x^{-4}y^6 = x^{-8}y^{12} = \frac{y^{12}}{x^8}$$

(b)

$$\frac{(-a^2)(-b)^5}{(-c^3)^2} - \frac{(-a)^2(-b^5)}{(-c^2)^3} = ?$$

**Lösung:**

$$\frac{(-a^2)(-b)^5}{(-c^3)^2} - \frac{(-a)^2(-b^5)}{(-c^2)^3} = \frac{a^2b^5}{c^6} - \frac{a^2b^5}{c^6} = 0$$

(c)

$$\left(-(-a^{-1})^2\right)^{-3} = ?$$

**Lösung:**

$$\begin{aligned} \left(-(-a^{-1})^2\right)^{-3} &= \left(-\left(-\frac{1}{a}\right)^2\right)^{-3} \\ &= \left(-\frac{1}{a^2}\right)^{-3} \\ &= (-1)^{-3} \cdot \left(\frac{1}{a^2}\right)^{-3} \\ &= -\frac{1}{a^{2 \cdot (-3)}} \\ &= -\frac{1}{a^{-6}} \\ &= -a^6 \end{aligned}$$

(d)

$$(-x^{-4})^{-3} (xy^5)^{-2} (-xy^2)^4 = ?$$

**Lösung:**

$$\begin{aligned} (-x^{-4})^{-3} (xy^5)^{-2} (-xy^2)^4 &= -x^{12}x^{-2}y^{-10}x^4y^8 \\ &= -x^{12-2+4}y^{-10+8} \\ &= -\frac{x^{14}}{y^2} \end{aligned}$$

(e)

$$\left(\frac{a^{-3}b}{x^{-2}y^{-3}}\right)^3 : \left(\frac{x^{-3}y}{ab}\right)^{-5} \frac{(x^{-1}y)^{-9}}{(a^{-7}b^{-1})^2} = ?$$

**Lösung:**

$$\begin{aligned} \left(\frac{a^{-3}b}{x^{-2}y^{-3}}\right)^3 &: \left(\frac{x^{-3}y}{ab}\right)^{-5} \frac{(x^{-1}y)^{-9}}{(a^{-7}b^{-1})^2} = \frac{a^{-9}b^3}{x^{-6}y^{-9}} \frac{x^{-15}y^5}{a^5b^5} \frac{x^9y^{-9}}{a^{-14}b^{-2}} \\ &= \frac{a^{14}b^2b^3x^6x^9y^9y^5}{a^9a^5b^5x^{15}y^9} \\ &= a^{14-9-5}b^{2+3-5}x^{6+9-15}y^{9+5-9} \\ &= a^0b^0x^0y^5 \\ &= y^5 \end{aligned}$$

(f)

$$-2^{-2} (-s^{-2}t)^{-3} (-st)^{-2} = ?$$

**Lösung:**

$$\begin{aligned} -2^{-2} (-s^{-2}t)^{-3} (-st)^{-2} &= -\frac{1}{2^2} \left(-\frac{t}{s^2}\right)^{-3} (-1)^2 \frac{1}{(st)^2} \\ &= -\frac{1}{2^2} \left(-\frac{t^{-3}}{s^{-6}}\right) \frac{1}{s^2t^2} \\ &= \frac{1}{2^2} \frac{s^6}{t^3} \frac{1}{s^2t^2} \\ &= \frac{s^4}{2^2t^5} \\ &= \frac{s^4}{4t^5} \end{aligned}$$

(g)

$$\left(\frac{2a^2}{3}\right)^{-3} \left(\frac{6}{5a^3}\right)^2 = ?$$

**Lösung:**

$$\begin{aligned} \left(\frac{2a^2}{3}\right)^{-3} \left(\frac{6}{5a^3}\right)^2 &= \frac{2^{-3}a^{-6}}{3^{-3}} \frac{6^2}{5^2a^6} \\ &= \frac{3^3}{2^3a^6} \frac{6^2}{5^2a^6} \\ &= \frac{3^36^2}{2^35^2a^{12}} \\ &= \frac{27 * 36}{8 * 25 * a^{12}} \\ &= \frac{243}{50a^{12}} \end{aligned}$$

(h)

$$(-2x^{-2})^3 (3x^{-2})^{-3} = ?$$

**Lösung:**

$$\begin{aligned} (-2x^{-2})^3 (3x^{-2})^{-3} &= (-1)^3 2^3 (x^{-2})^3 \frac{1}{(3x^{-2})^3} \\ &= \frac{-8x^{-6}}{3^3 x^{-6}} \\ &= -\frac{8}{27} \end{aligned}$$

(i)

$$\frac{(-3ab^2)^{-3}}{(2a^2b^{-3})^2} = ?$$

**Lösung:**

$$\begin{aligned} \frac{(-3ab^2)^{-3}}{(2a^2b^{-3})^2} &= \frac{(-1)^{-3} (3)^{-3} (a)^{-3} (b)^{-6}}{2^2 a^4 b^{-6}} \\ &= -\frac{1}{2^2 a^4 3^3 a^3} \\ &= -\frac{1}{2^2 3^3 a^7} \\ &= -\frac{1}{108 a^7} \end{aligned}$$

**2. Aufgabe**

Vereinfachen Sie und schreiben Sie das Ergebnis mit einer Wurzel:

(a)

$$\sqrt[7]{x^3} \sqrt[3]{x^7} = ?$$

**Lösung:**

$$\sqrt[7]{x^3} \sqrt[3]{x^7} = x^{\frac{3}{7}} x^{\frac{7}{3}} = x^{\frac{3}{7} + \frac{7}{3}} = x^{\frac{9+49}{21}} = \sqrt[21]{x^{58}}$$

(b)

$$\frac{\sqrt[3]{x^2} \sqrt[5]{x^3} x^2}{x \sqrt[15]{x} \sqrt[4]{x}} = ?$$

**Lösung:**

$$\begin{aligned} \frac{\sqrt[3]{x^2} \sqrt[5]{x^3} x^2}{x \sqrt[15]{x} \sqrt[4]{x}} &= \frac{x^{\frac{2}{3}} x^{\frac{3}{5}} x^2}{x x^{\frac{1}{15}} x^{\frac{1}{4}}} \\ &= x^{\frac{2}{3} + \frac{3}{5} + 2 - 1 - \frac{1}{15} - \frac{1}{4}} \\ &= x^{\frac{40+36+120-60-4-15}{60}} \\ &= x^{\frac{117}{60}} \\ &= x^{\frac{39}{20}} \\ &= \sqrt[20]{x^{39}} \end{aligned}$$

(c)

$$\sqrt{\frac{a}{1-a}} \sqrt[3]{\frac{(1-a)^2}{1+a}} \sqrt[4]{\frac{(1+a)^3}{a}} = ?$$

**Lösung:**

$$\begin{aligned} \sqrt{\frac{a}{1-a}} \sqrt[3]{\frac{(1-a)^2}{1+a}} \sqrt[4]{\frac{(1+a)^3}{a}} &= a^{\frac{1}{2}} (1-a)^{-\frac{1}{2}} (1-a)^{\frac{2}{3}} (1+a)^{-\frac{1}{3}} (1+a)^{\frac{3}{4}} a^{-\frac{1}{4}} \\ &= a^{\frac{1}{2}-\frac{1}{4}} (1-a)^{\frac{2}{3}-\frac{1}{2}} (1+a)^{\frac{3}{4}-\frac{1}{3}} \\ &= a^{\frac{1}{4}} (1-a)^{\frac{1}{6}} (1+a)^{\frac{5}{12}} \\ &= a^{\frac{3}{12}} (1-a)^{\frac{2}{12}} (1+a)^{\frac{5}{12}} \\ &= \sqrt[12]{a^3 (1-a)^2 (1+a)^5} \end{aligned}$$

(d)

$$\sqrt{\frac{\sqrt{ab^2c}}{\sqrt{b}}} : \frac{1}{\sqrt{\sqrt{abc^2}}} = ?$$

**Lösung:**

$$\begin{aligned} \sqrt{\frac{\sqrt{ab^2c}}{\sqrt{b}}} : \frac{1}{\sqrt{\sqrt{abc^2}}} &= \left( \frac{\sqrt{ab^2c}}{\sqrt{b}} \right)^{\frac{1}{2}} (\sqrt{abc^2})^{\frac{1}{2}} \\ &= \left( \frac{a^{\frac{1}{2}} b^2 c}{b^{\frac{1}{2}}} \right)^{\frac{1}{2}} (a^{\frac{1}{2}} b c^2)^{\frac{1}{2}} \\ &= \left( \frac{a^{\frac{1}{4}} b c^{\frac{1}{2}}}{b^{\frac{1}{4}}} \right) (a^{\frac{1}{4}} b^{\frac{1}{2}} c) \\ &= a^{\frac{1}{4}+\frac{1}{4}} b^{1-\frac{1}{4}+\frac{1}{2}} c^{\frac{1}{2}+1} \\ &= a^{\frac{1}{2}} b^{\frac{5}{4}} c^{\frac{3}{2}} \\ &= \sqrt{a} \sqrt[4]{b^5} \sqrt{c^3} \\ &= \sqrt[4]{a^2 b^5 c^6} \end{aligned}$$

(e)

$$\frac{\sqrt[3]{x^2 - 2x + 1}}{\sqrt[3]{x - 1}}$$

**Lösung:**

$$\frac{\sqrt[3]{x^2 - 2x + 1}}{\sqrt[3]{x - 1}} = \frac{\sqrt[3]{(x-1)^2}}{\sqrt[3]{x-1}} = \frac{(x-1)^{\frac{2}{3}}}{(x-1)^{\frac{1}{3}}} = (x-1)^{\frac{1}{3}} = \sqrt[3]{x-1}$$

(f)

$$\sqrt[5]{-\left(-\left(-\frac{1}{a}\right)^{-3}\right)^{-4}} = ?$$

**Lösung:**

$$\begin{aligned}
\dots &= \sqrt[5]{-((-(-a^{-1})^{-3})^{-4})} \\
&= \sqrt[5]{-((-(-1)^{-3}a^{-1 \cdot (-3)})^{-4})} \\
&= \sqrt[5]{-(a^3)^{-4}} \\
&= \sqrt[5]{-a^{3 \cdot (-4)}} \\
&= \sqrt[5]{-1} \cdot \sqrt[5]{\frac{1}{a^{12}}} \\
&= -\frac{1}{\sqrt[5]{a^{12}}}
\end{aligned}$$

(g)

$$\frac{\sqrt{a^2 - x^2}}{a^2 - x^2} + \frac{x^2}{\sqrt{(a^2 - x^2)^3}} = ?$$

**Lösung:**

$$\begin{aligned}
\frac{\sqrt{a^2 - x^2}}{a^2 - x^2} + \frac{x^2}{\sqrt{(a^2 - x^2)^3}} &= \frac{(a^2 - x^2)^{\frac{1}{2}}}{a^2 - x^2} + \frac{x^2}{(a^2 - x^2)^{\frac{3}{2}}} \\
&= \frac{(a^2 - x^2)^{\frac{1}{2}} (a^2 - x^2)^{\frac{1}{2}}}{(a^2 - x^2)(a^2 - x^2)^{\frac{1}{2}}} + \frac{x^2}{(a^2 - x^2)^{\frac{3}{2}}} \\
&= \frac{(a^2 - x^2) + x^2}{(a^2 - x^2)^{\frac{3}{2}}} \\
&= \frac{a^2}{\sqrt{(a^2 - x^2)^3}} = \sqrt{\frac{a^4}{(a^2 - x^2)^3}}
\end{aligned}$$

**3. Aufgabe**

Ordnen Sie die folgenden Terme nach aufsteigender Grösse (ohne Taschenrechner und mit Begründung!):

(a)

$$9^{12}, 27^7, 81^2, 3^{10}, 243^3, 3^{(3^3)}$$

**Lösung:**

Um die Terme Vergleichen zu können, schreiben wir alle Werte mit der gleichen Basis ( $b = 3$ ):

$$\begin{aligned}
9^{12} &= (3^2)^{12} = 3^{24} \\
27^7 &= (3^3)^7 = 3^{21} \\
81^2 &= (3^4)^2 = 3^8 \\
3^{10} &= 3^{10} \\
243^3 &= (3^5)^3 = 3^{15} \\
3^{(3^3)} &= 3^{27}
\end{aligned}$$

Gesuchte Reihenfolge (aufsteigend):

$$81^2 = 3^8 < 3^{10} < 243^3 = 3^{15} < 27^7 = 3^{21} < 9^{12} = 3^{24} < 3^{(3^3)} = 3^{27}$$

(b)

$$\sqrt{22^{222}}, \sqrt[22]{22^{22}}, \sqrt{22^{22}}, \sqrt[2]{\sqrt[22]{22^{222}}}$$

**Lösung:**

Darstellung als Potenzen zur Basis 2:

•  $\sqrt{22^{222}} :$

$$\log_2 \sqrt{22^{222}} = \frac{222}{2} \log_2 22$$

$$2^{444} < \sqrt{22^{222}} < 2^{555}$$

•  $\sqrt[22]{22^{22}} :$

$$\log_2 \sqrt[22]{22^{22}} = \frac{2}{22} \log_2 222$$

$$2^{\frac{7}{11}} < \sqrt[22]{22^{22}} < 2^{\frac{8}{11}}$$

•  $\sqrt{22^{222}} :$

$$\log_2 \sqrt{22^{222}} = \frac{22}{2} \log_2 222$$

$$2^{77} < \sqrt{22^{222}} < 2^{88}$$

•  $\sqrt[2]{\sqrt[22]{22^{222}}} :$

$$\log_2 \sqrt[2]{\sqrt[22]{22^{222}}} = \frac{222}{22 * 2} \log_2 2$$

$$\sqrt[2]{\sqrt[22]{22^{222}}} = 2^{\frac{101}{2}}$$

Somit findet man:

$$2^{\frac{7}{11}} < \sqrt[22]{22^{22}} < 2^{\frac{8}{11}} < \sqrt[2]{\sqrt[22]{22^{222}}} = 2^{\frac{101}{2}} < 2^{77} < \sqrt{22^{222}} < 2^{88} < 2^{444} < \sqrt{22^{222}} < 2^{555}$$

#### 4. Aufgabe

Bestimmen Sie die folgenden Logarithmen von Hand:

(a)

$$\log_5 \frac{1}{125} = ?$$

**Lösung:**

$$\log_5 \frac{1}{125} = \log_5 \frac{1}{5^3} = \log_5 5^{-3} = -3$$

(b)

$$\log_{10} \sqrt[3]{100} = ?$$

**Lösung:**

$$\log_{10} \sqrt[3]{100} = \log_{10} \sqrt[3]{10^2} = \log_{10} 10^{\frac{2}{3}} = \frac{2}{3}$$

(c)

$$\log_{16} 2 = ?$$

**Lösung:**

$$\log_{16} 2 = \log_{16} 2^{\frac{4}{4}} = \log_{16} 16^{\frac{1}{4}} = \frac{1}{4}$$

(d)

$$\log_{\frac{1}{4}} \sqrt{8} = ?$$

**Lösung:**

$$\log_{\frac{1}{4}} \sqrt{8} = \log_{\frac{1}{4}} \sqrt{\sqrt{64}} = \log_{\frac{1}{4}} \sqrt[4]{64} = \log_{\frac{1}{4}} 4^{\frac{3}{4}} = \log_{\frac{1}{4}} 4^{\frac{3}{4}} = \log_{\frac{1}{4}} \left(\frac{1}{4}\right)^{-\frac{3}{4}} = -\frac{3}{4}$$

(e)

$$\log_9 \frac{1}{\sqrt{27}} = ?$$

**Lösung:**

$$\log_9 \frac{1}{\sqrt{27}} = \log_9 \frac{1}{\sqrt{\sqrt{729}}} = \log_9 \frac{1}{\sqrt[4]{9^3}} = \log_9 \frac{1}{9^{\frac{3}{4}}} = \log_9 9^{-\frac{3}{4}} = -\frac{3}{4}$$

(f)

$$\log_8 (\log_4 (\log_2 (2^{16}))) = ?$$

**Lösung:**

$$\begin{aligned} \log_8 (\log_4 (\log_2 (2^{16}))) &= \log_8 (\log_4 (16)) \\ &= \log_8 (\log_4 (4^2)) \\ &= \log_8 (2) \\ &= \log_8 \left(8^{\frac{1}{3}}\right) \\ &= \frac{1}{3} \end{aligned}$$

(g)

$$\log_9 \frac{\sqrt{3}}{27} = ?$$

**Lösung:**

$$\log_9 \frac{\sqrt{3}}{27} = \log_9 \frac{\sqrt{3}}{3^3} = \log_9 3^{\frac{1}{2}-3} = \log_9 3^{-\frac{5}{2}} = \log_9 \sqrt{9}^{-\frac{5}{2}} = \log_9 9^{-\frac{5}{4}} = -\frac{5}{4}$$

(h)

$$\log_{\sqrt{2}} 8 = ?$$

**Lösung:**

$$\log_{\sqrt{2}} 8 = \log_{\sqrt{2}} 2^3 = \log_{\sqrt{2}} \left(\sqrt{2}^2\right)^3 = \log_{\sqrt{2}} \sqrt{2}^6 = 6$$

(i)

$$\log_8 \frac{1}{\sqrt[3]{4}} = ?$$

**Lösung:**

$$\log_8 \frac{1}{\sqrt[3]{4}} = \log_8 \frac{1}{4^{\frac{1}{3}}} = \log_8 4^{-\frac{1}{3}} = \log_8 \sqrt[3]{64}^{-\frac{1}{3}} = \log_8 64^{-\frac{1}{9}} = \log_8 (8^2)^{-\frac{1}{9}} = -\frac{2}{9} \log_8 8 = -\frac{2}{9}$$

(j)

$$\log_{\frac{1}{9}} 243 = ?$$

**Lösung:**

$$\log_{\frac{1}{9}} 243 = \log_{\frac{1}{9}} 3^5 = \log_{\frac{1}{9}} \sqrt{9}^5 = \log_{\frac{1}{9}} 9^{\frac{5}{2}} = \log_{\frac{1}{9}} \left( \left( \frac{1}{9} \right)^{-1} \right)^{\frac{5}{2}} = -\frac{5}{2} \log_{\frac{1}{9}} \frac{1}{9} = -\frac{5}{2}$$

(k)

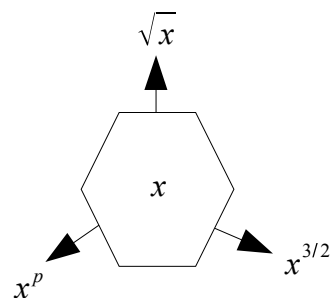
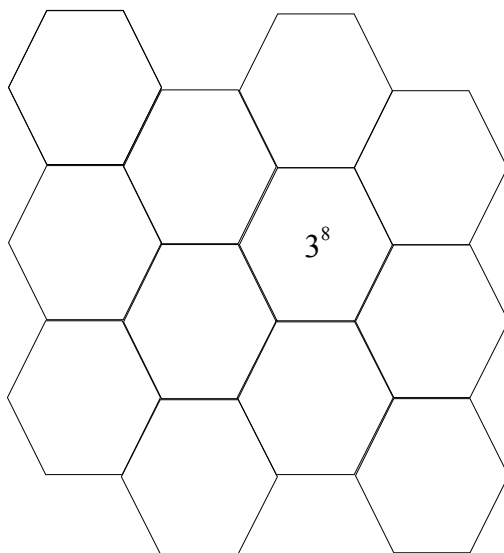
$$\log_4 \sqrt{8} = ?$$

**Lösung:**

$$\log_4 \sqrt{8} = \log_4 8^{\frac{1}{2}} = \log_4 \sqrt{64}^{\frac{1}{2}} = \log_4 (4^3)^{\frac{1}{4}} = \frac{3}{4} \log_4 4 = \frac{3}{4}$$

**5. Aufgabe**

Füllen Sie untenstehendes Wabenmuster aus und geben Sie den Wert für  $p$  an:

**Lösung:**

- Nach Oben:

$$\sqrt{3^8} = 3^4$$

- Nach Unten:

$$(3^8)^2 = 3^{16}$$

- Nach Rechts-Unten:

$$(3^8)^{\frac{3}{2}} = 3^{12}$$

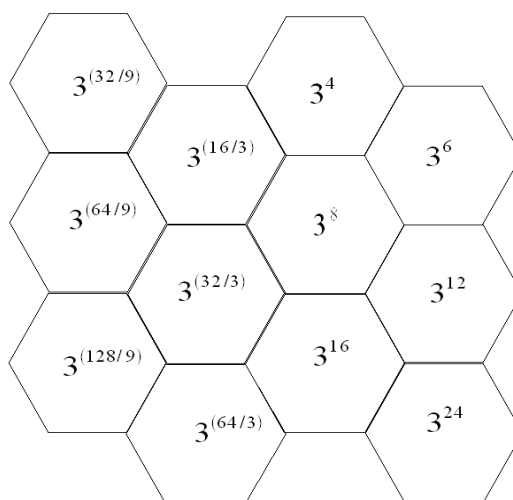
- Nach Rechts-Oben:

$$\sqrt{3^{12}} = 3^6$$

- Parameter (nach Rechts-Oben):

$$\begin{aligned} (3^6)^p &= 3^8 \\ 6p &= 8 \\ p &= \frac{4}{3} \end{aligned}$$

- Lösung:



## 6. Aufgabe

- (a) Fassen Sie zu einem Logarithmus zusammen:

•

$$\frac{1}{2} [\ln(a+b) + \ln(a-b) + \ln(a^2+b^2) + \ln(a^4+b^4)] = ?$$

**Lösung:**

$$\begin{aligned} \dots &= \frac{1}{2} [\ln((a+b)(a-b)(a^2+b^2)(a^4+b^4))] \\ &= \frac{1}{2} [\ln((a^2-b^2)(a^2+b^2)(a^4+b^4))] \\ &= \frac{1}{2} [\ln((a^4-b^4)(a^4+b^4))] \\ &= \frac{1}{2} [\ln(a^8-b^8)] \\ &= \ln \sqrt{a^8-b^8} \end{aligned}$$

•

$$\frac{1}{4} \lg((x-y)^2) - \frac{1}{2} \lg(x^2-y^2) + \frac{1}{2} \lg(x+y) = ?$$

**Lösung:**

$$\begin{aligned}
\dots &= \lg(x-y)^{\frac{1}{2}} - \lg(x^2-y^2)^{\frac{1}{2}} + \lg(x+y)^{\frac{1}{2}} \\
&= \lg \frac{(x-y)^{\frac{1}{2}} (x+y)^{\frac{1}{2}}}{(x^2-y^2)^{\frac{1}{2}}} \\
&= \lg \left( \frac{(x-y)(x+y)}{(x^2-y^2)} \right)^{\frac{1}{2}} \\
&= \lg(1)^{\frac{1}{2}} = 0
\end{aligned}$$

•

$$\frac{1}{3} \lg(a) - \frac{1}{4} \lg(b) + 2 \lg(ab) = ?$$

**Lösung:**

$$\begin{aligned}
\frac{1}{3} \lg(a) - \frac{1}{4} \lg(b) + 2 \lg(ab) &= \lg(a^{\frac{1}{3}}) - \lg(b^{\frac{1}{4}}) + \lg(a^2 b^2) \\
&= \lg \frac{a^{\frac{1}{3}} a^2 b^2}{b^{\frac{1}{4}}} \\
&= \lg(a^{\frac{7}{3}} b^{\frac{7}{4}})
\end{aligned}$$

•

$$\lg(a) - 3 \left( \frac{1}{2} \lg(b-c) - \lg(b+c) \right) = ?$$

**Lösung:**

$$\begin{aligned}
\lg(a) - 3 \left( \frac{1}{2} \lg(b-c) - \lg(b+c) \right) &= \lg(a) - \frac{3}{2} \lg(b-c) + 3 \lg(b+c) \\
&= \lg(a) - \lg(b-c)^{\frac{3}{2}} + \lg(b+c)^3 \\
&= \lg \left( \frac{a(b+c)^3}{(\sqrt{b-c})^3} \right)
\end{aligned}$$

(b) Schreiben Sie den Term als Summe/Differenz von Logarithmen:

•

$$\log_4 \sqrt{\frac{16a^4}{\sqrt{a^4-b^4}}} = ?$$

**Lösung:**

$$\begin{aligned}
\log_4 \sqrt{\frac{16a^4}{\sqrt{a^4-b^4}}} &= \frac{1}{2} \log_4 \frac{16a^4}{\sqrt{a^4-b^4}} \\
&= \frac{1}{2} \left( \log_4 16a^4 - \log_4 \sqrt{a^4-b^4} \right) \\
&= \frac{1}{2} \left( \log_4 16 + \log_4 a^4 - \frac{1}{2} \log_4 (a^4-b^4) \right) \\
&= 1 + 2 \log_4 a - \frac{1}{4} \log_4 ((a^2+b^2)(a-b)(a+b)) \\
&= 1 + 2 \log_4 a - \frac{1}{4} \log_4 (a^2+b^2) - \frac{1}{4} \log_4 (a-b) - \frac{1}{4} \log_4 (a+b)
\end{aligned}$$

•

$$\lg \left( a^2 \left( b^{-3} \frac{\sqrt{a}}{\sqrt{a^{-3}}} \right)^{-2} \right) = ?$$

**Lösung:**

$$\begin{aligned} \lg \left( a^2 \left( b^{-3} \frac{\sqrt{a}}{\sqrt{a^{-3}}} \right)^{-2} \right) &= \lg(a^2) + \lg \left( b^{-3} \frac{\sqrt{a}}{\sqrt{a^{-3}}} \right)^{-2} \\ &= 2 \lg(a) - 2 \lg \left( b^{-3} \frac{\sqrt{a}}{\sqrt{a^{-3}}} \right) \\ &= 2 \lg(a) - 2 \left( \lg(b^{-3}) + \lg(\sqrt{a}) - \lg(\sqrt{a^{-3}}) \right) \\ &= 2 \lg(a) - 2 \left( -3 \lg(b) + \frac{1}{2} \lg(a) - \left( -\frac{3}{2} \right) \lg(a) \right) \\ &= 2 \lg(a) + 6 \lg(b) - \lg(a) - 3 \lg(a) \\ &= -2 \lg(a) + 6 \lg(b) \end{aligned}$$

•

$$\lg \left( \frac{x^2}{y^3} \sqrt[4]{\frac{a^3}{3b}} \right) = ?$$

**Lösung:**

$$\begin{aligned} \lg \left( \frac{x^2}{y^3} \sqrt[4]{\frac{a^3}{3b}} \right) &= \lg(x^2) - \lg(y^3) + \lg \left( \sqrt[4]{\frac{a^3}{3b}} \right) \\ &= 2 \lg(x) - 3 \lg(y) + \frac{1}{4} \lg \left( \frac{a^3}{3b} \right) \\ &= 2 \lg(x) - 3 \lg(y) + \frac{1}{4} (\lg(a^3) - \lg(3) - \lg(b)) \\ &= 2 \lg(x) - 3 \lg(y) + \frac{3}{4} \lg(a) - \frac{1}{4} \lg(3) - \frac{1}{4} \lg(b) \end{aligned}$$

•

$$\lg \left( \frac{x}{\sqrt{x-y}} \sqrt{\frac{x+y}{x^3}} \right) = ?$$

**Lösung:**

$$\begin{aligned} \lg \left( \frac{x}{\sqrt{x-y}} \sqrt{\frac{x+y}{x^3}} \right) &= \lg(x) - \frac{1}{2} \lg(x-y) + \frac{1}{2} \lg(x+y) - \frac{3}{2} \lg(x) \\ &= \frac{1}{2} \lg(x+y) - \frac{1}{2} \lg(x-y) - \frac{1}{2} \lg(x) \end{aligned}$$

**7. Aufgabe**

Schreiben Sie die folgenden Grössen in SI-Einheiten und wissenschaftlicher Schreibweise:

(a)

$$243.865 \text{Lichtjahre} = ? [m]$$

**Lösung:**

$$\begin{aligned} 243.865 \text{Lichtjahre} &= 243.865 \cdot 365 \cdot 24 \cdot 60 \cdot 60 \cdot 3000000000 m \\ &= 2.3072 \cdot 10^{18} m \end{aligned}$$

(b) ( $1eV = 1.60217733 \cdot 10^{-19} \frac{kg \cdot m}{s^2}$ )

$$0.0000456 eV = ? \left[ \frac{kg \cdot m}{s^2} \right]$$

**Lösung:**

$$\begin{aligned} 0.0000456 eV &= 0.0000456 \cdot 1.60217733 \cdot 10^{-19} \frac{kg \cdot m}{s^2} \\ &= 7.3059 \cdot 10^{-24} \frac{kg \cdot m}{s^2} \end{aligned}$$